

Written exam December 17, 2015.

Course name: Network Optimization

Course no: 42115

Rules: You are allowed to use your books, notes, slides, computer (but no internet access). You can write in Danish or English.

Assignment: The assignment consists of 12 problems. In some of the problems it is written what an answer is expected to contain. Notice that all figures can be found on the last pages of this assignment. You can use them in your answer by drawing on the figure.

Points: Small problems give 2 points, medium problems give 3 points, large problems 4 points. A total of 32 points can be obtained.

Critical path problem

The below table shows the precedence relations and durations for a nine activity project. The task is to schedule all activities so as to minimize the overall completion time.

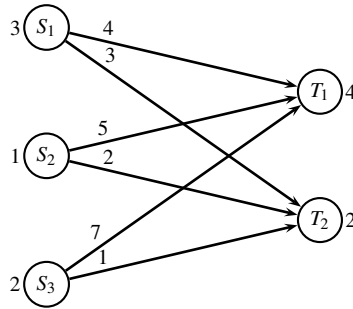
Activity	Description	Predecessors	Duration
A	Site clearing		4
B	Removal of trees		3
C	General excavation	A	8
D	Grading general area	A	7
E	Excavation for trenches	B,C	9
F	Placing formwork and reinforcement for concrete	B,C	12
G	Installing sewer lines	D,E	2
H	Installing other utilities	D,E	5
I	Pouring concrete	F,G	6

Problem 1: (medium) For each activity, find the earliest starting time and the latest starting time. ■

Problem 2: (small) Mark the critical path. ■

Fixed charge transportation problem

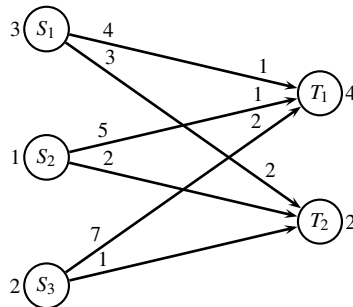
Three factories S_1, S_2, S_3 are transporting items to two customers T_1, T_2 . The *transportation costs* of sending one item is indicated on the graph below. The production at factories are respectively $p_1 = 3, p_2 = 1, p_3 = 2$, while the demands at the customers are $d_1 = 4, d_2 = 2$. There is a *fixed charge* of cost 5 associated with each transportation. The fixed charge is to be paid if one or more items are sent along an edge, however, the fixed charge is independant of the number of items sent.



Since fixed charge transportation problems are known to be difficult to solve, we use a Dantzig-Wolfe decomposed formulation, where we write up all patterns of sending the p_i items from node S_i . The cost of a pattern consists of the transportation costs, and the fixed charges for using edges.

Problem 3: (medium) Write up the Dantzig-Wolfe decomposed model. The first constraint should make sure that the d_1 items arrive at customer T_1 , and the second constraint should make sure that the d_2 items arrive at customer T_2 . ■

Since the Dantzig-Wolfe decomposed model may be quite large, it is solved through column generation. The initial three columns are chosen by solving the transportation problem by a greedy method, getting the following solution. Values at the start of an edge indicate the edge cost, while values at the end of an edge indicate the flow.



Problem 4: (small) Which three columns in the Dantzig-Wolfe formulation correspond to the greedy solution? ■

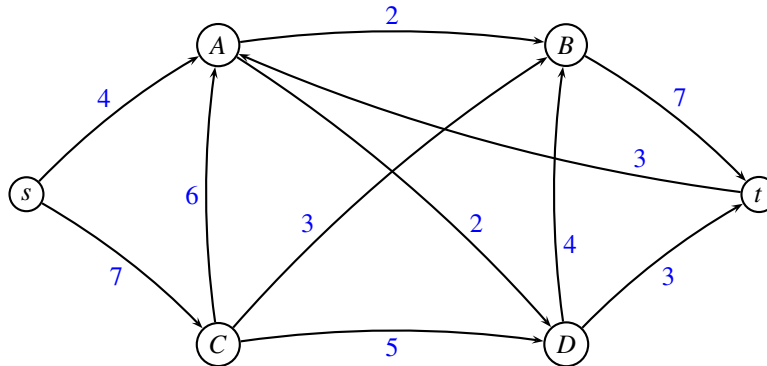
The restricted master problem is solved with only these three columns. Let u_i be the dual variables corresponding to the constraints saying that d_i items arrive at customer T_i . Let v_j be the dual variables corresponding to the constraints saying that only one pattern may be used for factory S_j .

The dual variables for the restricted master problem are $u_1 = 0$, $u_2 = 10$, $v_1 = 0$, $v_2 = 10$, $v_3 = 19$.

Problem 5: (small) Calculate the reduced costs of all patterns (which are not currently used in the restricted master problem). Which pattern should be added to the restricted master problem (and why?). ■

Maximum flow

Consider the following maximum flow problem where edge weights denote the capacity of an edge.



Problem 6: (medium) Solve the above maximum flow problem using an augmenting path algorithm. In each iteration use the augmenting path having *largest possible capacity*. Report which augmenting paths were used and their capacity. (You do not need to draw the residual graph in each iteration) ■

Problem 7: (small) Indicate a minimum cut, and report the capacity of the cut and the value of the flow. ■

Minimum cost flow problem

A caterer provides napkins on a 5-day schedule. The demand for each of the five days is:

Day	1	2	3	4	5
No. napkins	300	200	400	100	500

The caterer begins with no napkins, and to supply napkins on any particular day she has three options:

- Purchase new napkins at 8 kr per napkin
- Quick-Launder dirty napkins for the following day at 3 kr per napkin
- Slow-Launder dirty napkins for 2 days following at 1 kr per napkin

After the fifth day all napkins are discarded. How should the caterer go about providing napkins for the 5-day period?

Problem 8: (large) Formulate the problem as a minimum cost flow problem. It is sufficient to draw the graph, indicating demands on nodes, and costs/capacities on all edges. You do not need to solve the problem. ■

Arbitrage opportunity

A given day the exchange rates are as follows. Each row says how much you get for 1 unit of the given currency.

	USD	EUR	DKK	CHF	GBP
USD	1.00	.88	6.54	.96	.65
EUR	1.14	1.00	7.46	1.09	.74
DKK	.15	.13	1.00	.15	.10
CHF	1.05	.92	6.84	1.00	.68
GBP	1.54	1.35	10.05	1.47	1.00

Assume that you can convert 1 USD to 0.88 EUR, and then 0.88 EUR to some other currency, for finally converting it back to USD such that you end up having more than 1 USD. This is call an *arbitrage opportunity*.

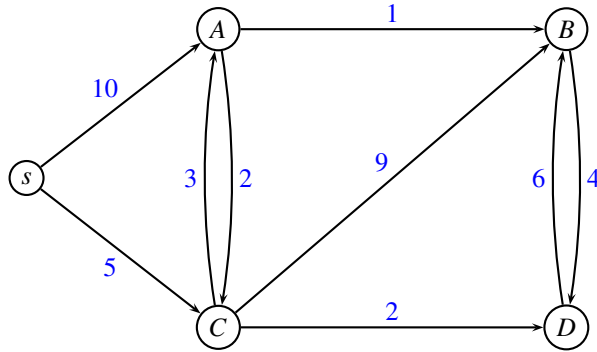
To be more formal, let e_{ij} be the exchange rate of currency i to currency j . Thus we are looking for a sequence of currencies $i, j, k, \dots, m, n$ such that $e_{ij} \cdot e_{jk} \cdots e_{mn} > 1$.

Problem 9: (large) Formulate the problem as a network problem, and say which algorithm you would use to solve the problem. You do not need to actually solve the above problem. The formulation should be in general terms, assuming that there are N currencies available. ■

Hint: Use the fact that $\log(e_{ij} \cdot e_{jk} \cdots e_{mn}) = \log(e_{ij}) + \log(e_{jk}) + \cdots + \log(e_{mn})$.

Shortest path

In the following network the edge weights denote the distance between the end nodes.



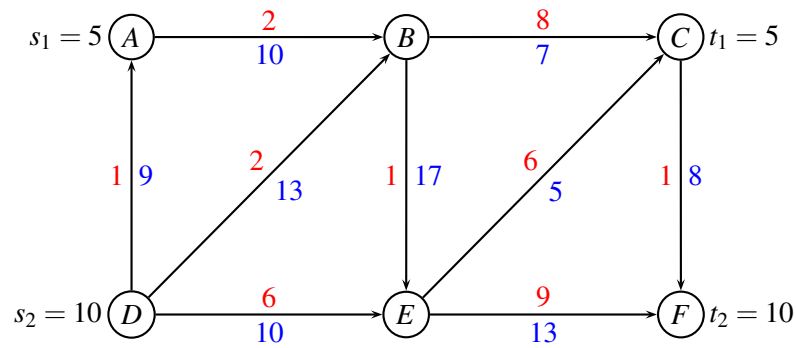
Problem 10: (medium) Use Dijkstra's algorithm to find a shortest path from s to all nodes. For every node report the shortest distance, and the shortest path. Give sufficient details so that the solution method can be followed. ■

Multi-commodity flow problem

An interesting variant of the multi-commodity flow problem is the one in which, in addition to having a capacity u_e for every edge e , we also have a capacity u_v for every node v . A flow x is feasible only if the flow out from a node v is less or equal to u_v for all nodes. The normal flow conservation constraints, and edge capacity constraints should still be satisfied.

Consider the following instance of the multi-commodity flow problem. Each edge e has a capacity u_e (blue), and a unit transportation cost c_e (red).

Assume that node B and E has a capacity $u_B = u_E = 12$, while all other nodes have unlimited capacity.

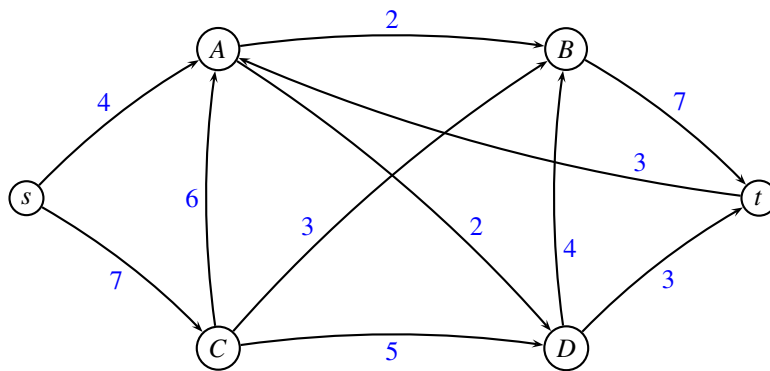


Problem 11: (small) Show how the problem can be transformed to a standard multi-commodity flow problem (where we only have capacities on edges). ■

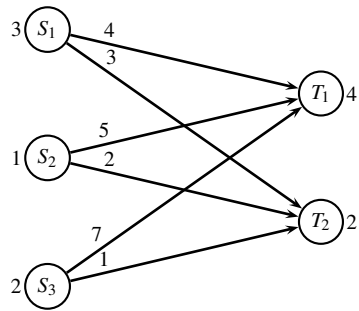
Problem 12: (small) Solve the resulting standard multi-commodity flow problem, assuming that all demands must be transported *unsplittable*. ■

_____end of assignment _____

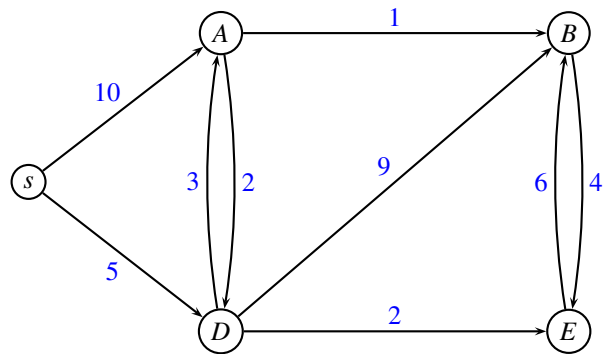
Maximum flow



Fixed charge transportation



Shortest path



Multi-commodity flow problem

